

PROJECT INITIALIZATION AND PLANNING PHASE

Date	27 June 2026
Team ID	
Project Title	Rising Waters
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to improve flood prediction using machine learning, enhancing disaster preparedness and reducing the impact of floods. It addresses the limitations of traditional forecasting methods by providing accurate, real-time flood risk predictions. Key features include a machine learning-based flood prediction model, weather data analysis, and real-time flood alerts.

PROJECT OVERVIEW	
Objective	The primary objective is to develop an intelligent flood prediction system using advanced machine learning techniques to accurately predict flood occurrences and provide early warnings.
Scope	The project focuses on collecting and analyzing historical weather, rainfall, river water level, and environmental data to predict flood risks and assist disaster management authorities.
PROBLEM STATEMENT	
Description	Existing flood prediction methods often lack accuracy and fail to provide timely warnings, resulting in property damage, economic losses, and risks to human lives.
Impact	Implementing an AI-based flood prediction system will improve forecasting accuracy, enable early warning systems, reduce disaster losses, and support effective emergency planning.
PROPOSED SOLUTION	
Approach	Develop a machine learning model that analyzes rainfall, temperature, humidity, river water levels, soil moisture, and historical flood data to predict flood occurrence.

Key Features	• Machine learning-based flood prediction model. • Real-time weather and river level monitoring. • Early flood warning notifications. • Interactive visualization of flood-prone areas. • Continuous model improvement using new environmental data.
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Resource Requirements

Resource Type	Description	Specification/Allocation
HARDWARE		
Computing Resources	CPU/GPU specifications	T4 GPU
Memory	RAM specifications	8 GB
Storage	Data storage for datasets and models	1 TB SSD
SOFTWARE		
Frameworks	Python frameworks	Flask
Libraries	Machine Learning & Data Processing Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, TensorFlow (optional), XGBoost.
Development Environment	IDE	Jupyter Notebook, PyCharm, VS Code
DATA		
Dataset	Historical weather and flood data	Kaggle Flood Prediction Dataset, NOAA Weather Dataset, IMD Rainfall Dataset (CSV format)